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INTRUSION DETECTION SYSTEM

Network Malicious Detection (Benign vs Malicious)

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All NUSW-NB15_1 Dataset shape: (699933, 43)

Dataset after drop duplicates: (639506, 43)

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Label (Benign and Malicious) counts:

Label

0 626310

1 13196

Name: count, dtype: int64

Features after cleaning: 42

Data split: Train=447654 (70%), Test=191852 (30%)

Training model...

Training time: 12.50 seconds

Cross-Validation (on training set)...

10-Fold CV Accuracy on Train: 0.9969 ± 0.0003

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PERFORMANCE SUMMARY

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Accuracy | Train: 99.72% | Test: 99.66%

Balanced Acc | Train: 99.84% | Test: 99.68%

Precision | Train: 99.75% | Test: 99.71%

Recall | Train: 99.72% | Test: 99.66%

F1 | Train: 99.73% | Test: 99.67%

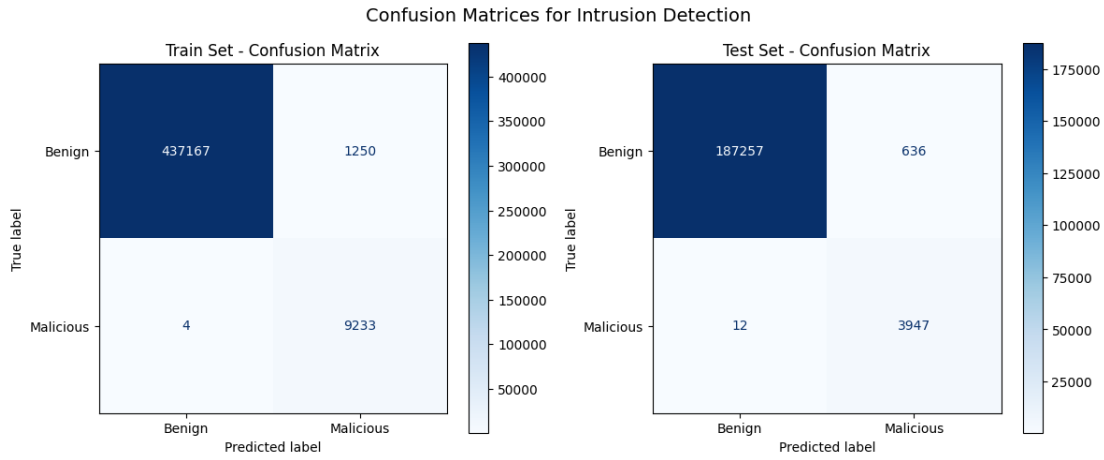
MCC | Train: 93.69% | Test: 92.50%

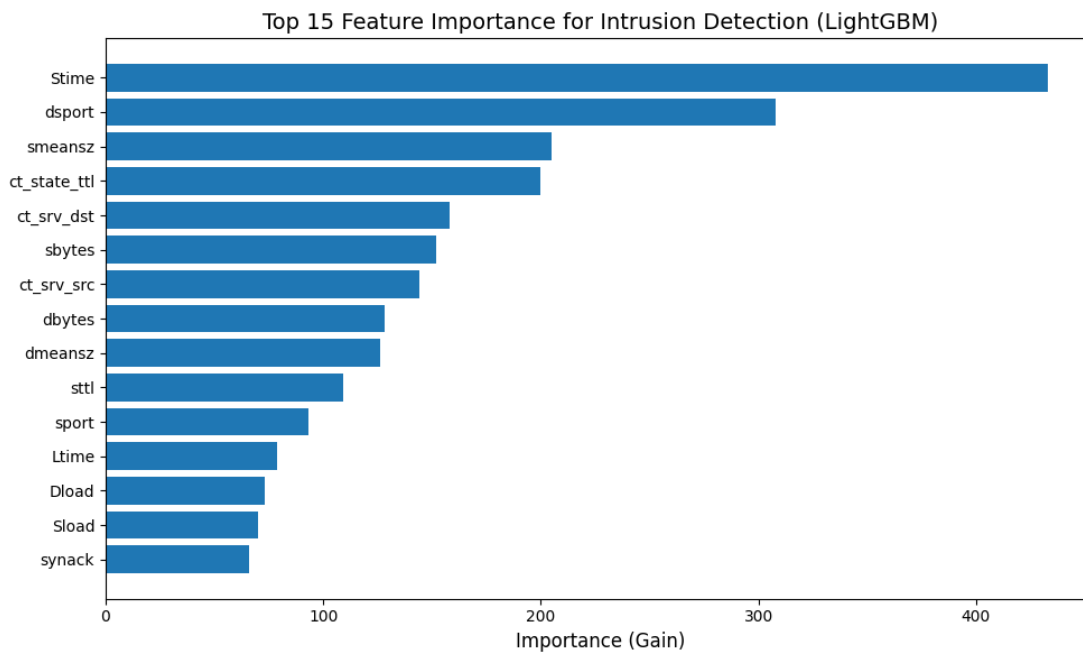
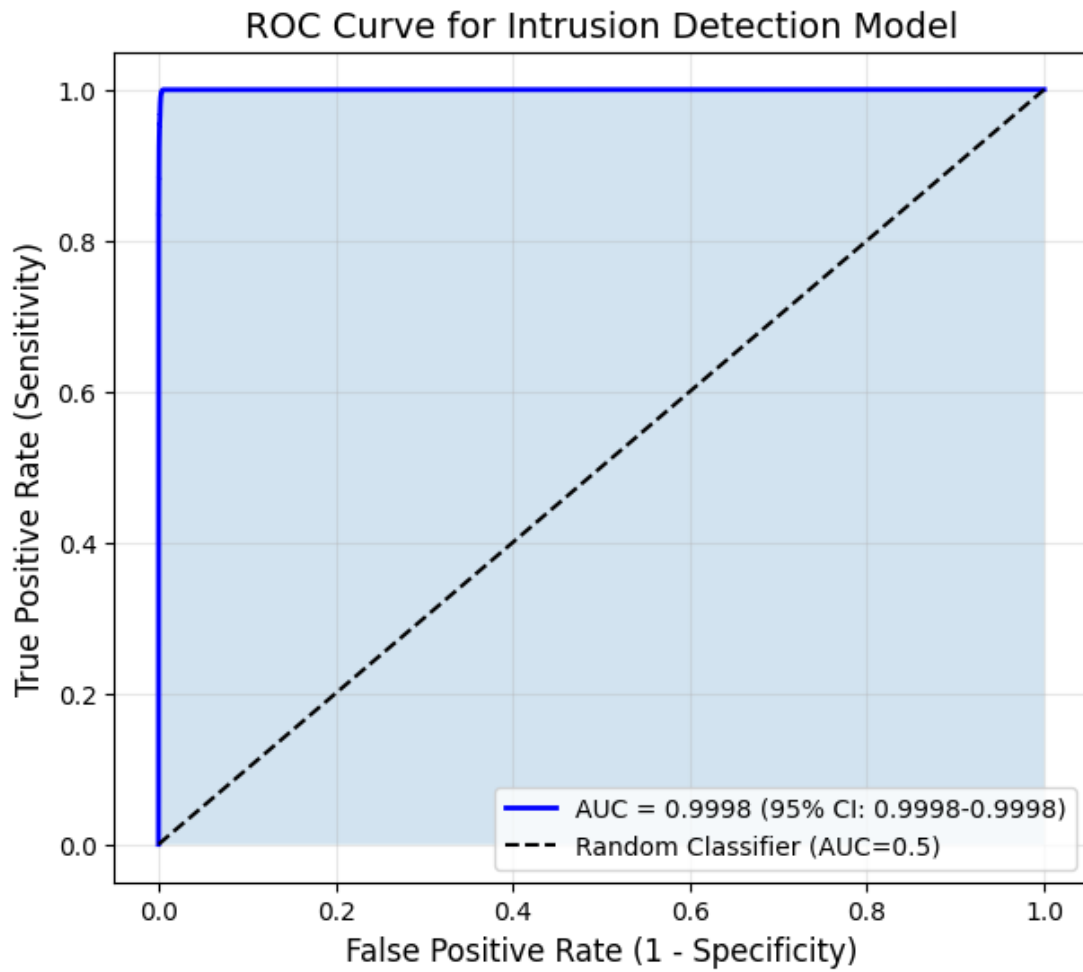
Kappa | Train: 93.50% | Test: 92.24%

Specificity | Train: 99.71% | Test: 99.66%

Sensitivity | Train: 99.96% | Test: 99.70%

AUC | Train: 0.00% | Test: 99.98%





Top 10 features:

Feature Importance

23	Stime	433
1	dsport	308
17	smeansz	205
31	ct_state_ttl	200
36	ct_srv_dst	158
3	sbytes	152
35	ct_srv_src	144
4	dbytes	128
18	dmeansz	126
5	sttl	109

SHAP analysis (sampling 100 test samples)...

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FEATURE IMPORTANCE BASED ON SHAP

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Feature	Strength	Impact %

ct_state_ttl	0.1664	49.32
sttl	0.1264	37.46
dsport	0.0078	2.31
dbytes	0.0055	1.62
Dload	0.0050	1.47
smeansz	0.0050	1.47
dmeansz	0.0049	1.46
Ltime	0.0048	1.42
ct_srv_dst	0.0022	0.64

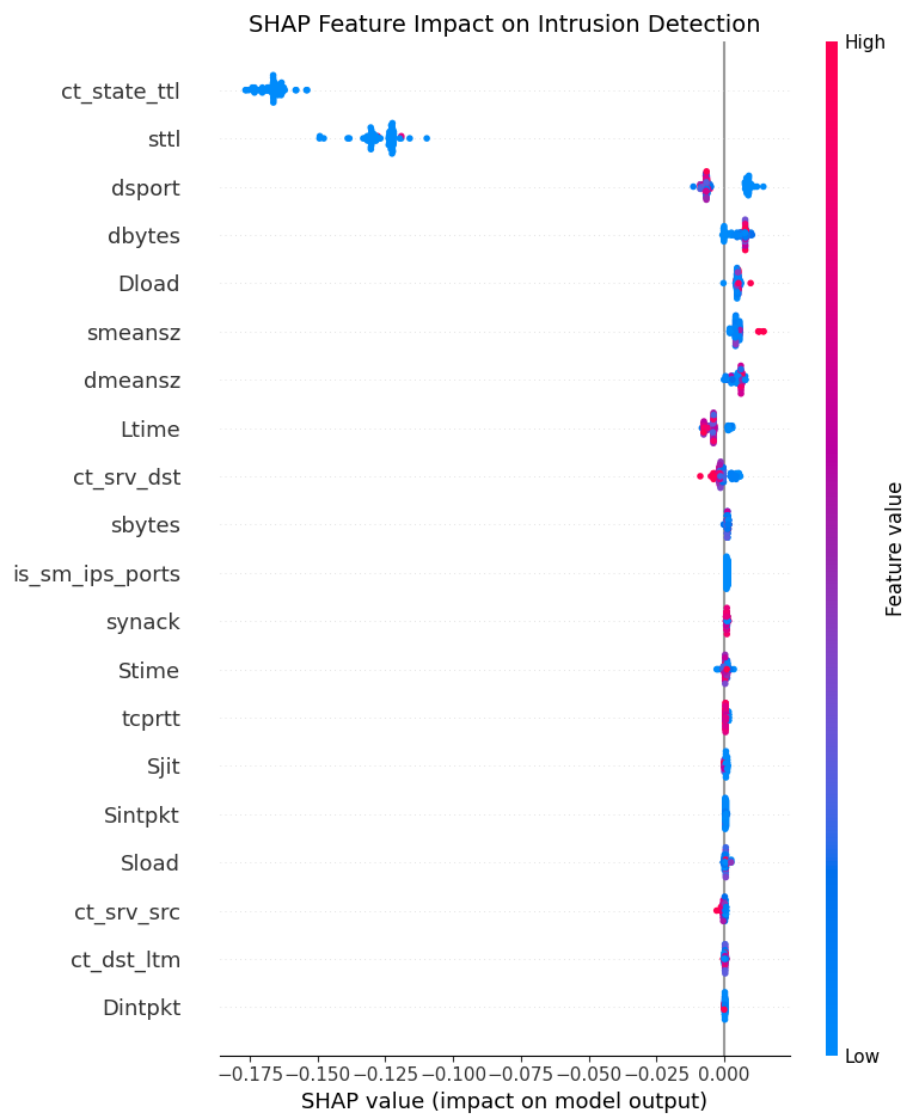
sbytes	0.0012	0.36
is_sm_ips_ports	0.0011	0.32
synack	0.0010	0.29
Stime	0.0008	0.23
tcprtt	0.0007	0.22
Sjit	0.0006	0.17

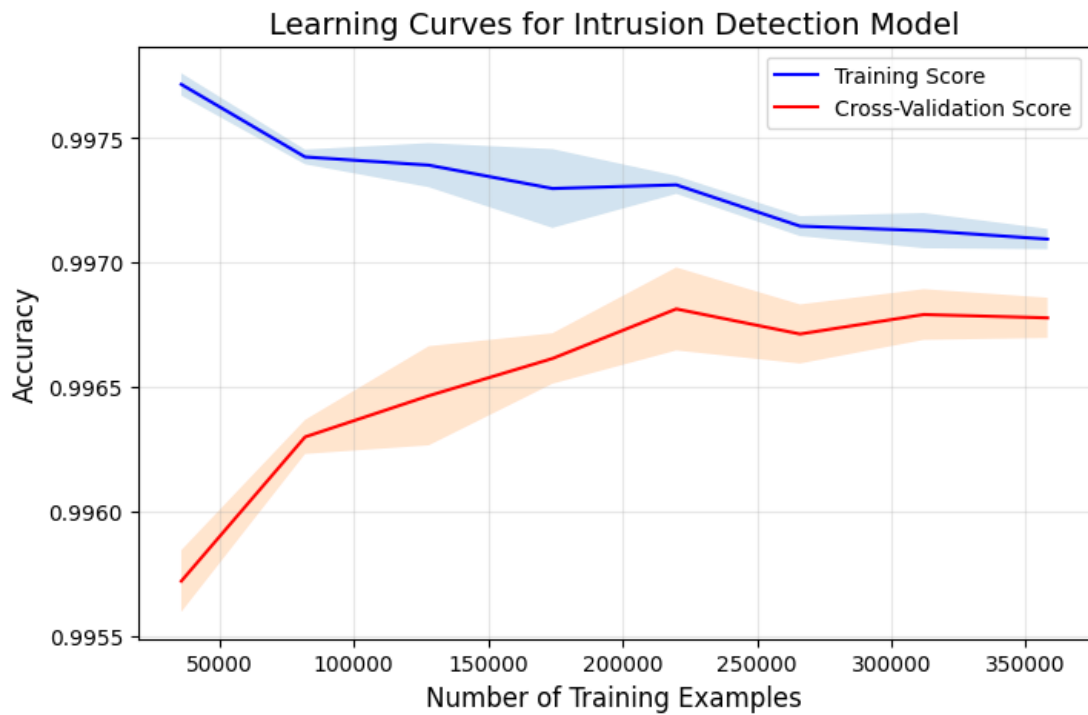
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Top feature: ct_state_ttl (Impact: 49.32%)

Second feature: sttl (Impact: 37.46%)

Third feature: dsport (Impact: 2.31%)

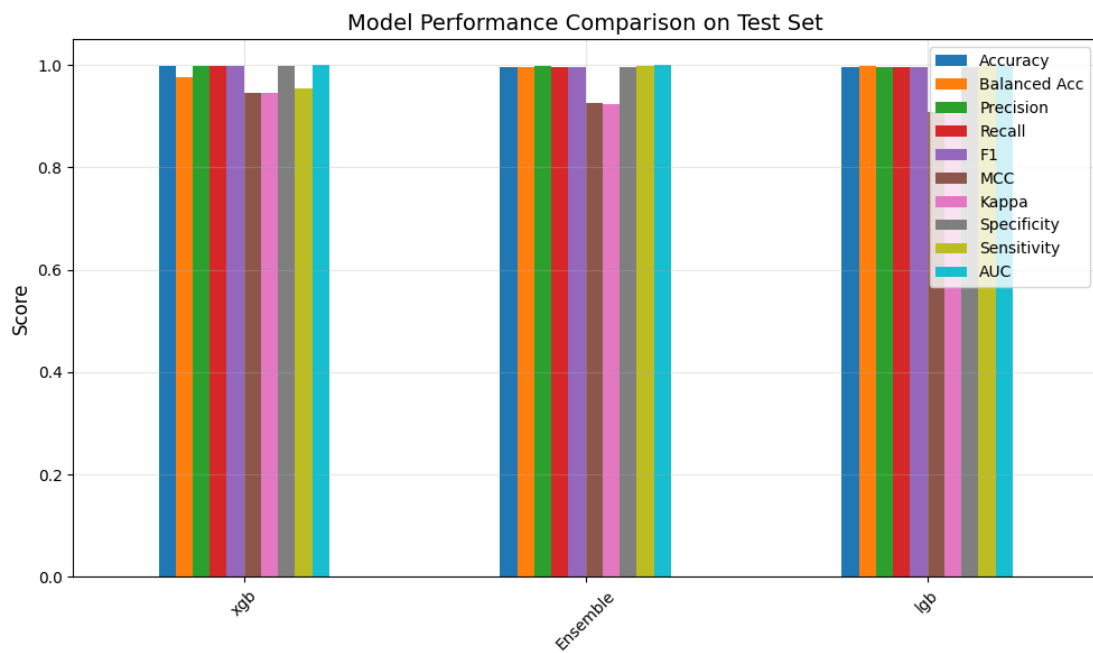




Comparing with individual models on test set...

	Accuracy	Balanced Acc	Precision	Recall	F1	MCC	Kappa \
xgb	0.9978	0.9760	0.9978	0.9978	0.9978	0.9463	0.9463
Ensemble	0.9966	0.9968	0.9971	0.9966	0.9967	0.9250	0.9224
lgb	0.9958	0.9972	0.9965	0.9958	0.9959	0.9086	0.9045

	Specificity	Sensitivity	AUC
xgb	0.9988	0.9533	0.9998
Ensemble	0.9966	0.9970	0.9998
lgb	0.9957	0.9987	0.9998



McNemar's test (Ensemble vs Logistic Regression): p-value = 0.000000

Result: Statistically significant difference ($p < 0.05$)

Model saved to fast_ids_model.pkl

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FINAL CLASSIFICATION REPORT (Test Set)

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	precision	recall	f1-score	support
Benign	1.00	1.00	1.00	187893
Malicious	0.86	1.00	0.92	3959
accuracy			1.00	191852
macro avg	0.93	1.00	0.96	191852
weighted avg	1.00	1.00	1.00	191852

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INTRUSION DETECTION SYSTEM - COMPLETED SUCCESSFULLY

Results saved in: ids_results/

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Sample predictions on test instances:

Correct | True: Benign | Predicted: Benign

Correct | True: Benign | Predicted: Benign

Correct | True: Benign | Predicted: Benign

Correct | True: Benign | Predicted: Benign

Correct | True: Benign | Predicted: Benign

Total execution time: 17.83 minutes

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